

ABSTRACT

The Smart Presence Monitoring System (SPMS) is a comprehensive web-based attendance management system designed specifically for educational institutions. The primary purpose of this system is to automate and simplify the attendance recording process for both teachers and students, thereby eliminating the inefficiencies associated with traditional manual attendance systems.

Traditional attendance systems rely on manual roll calls or paper registers, which consume valuable classroom time and are prone to human errors. The SPMS system provides a digital solution where teachers can create attendance sessions and students can mark their attendance using a secure session ID and PIN. This approach not only saves time but also prevents proxy attendance, ensuring the integrity of attendance records.

The system employs a modern three-tier architecture consisting of a frontend developed using HTML, CSS, and JavaScript, a backend developed using Node.js and Express.js, and a MongoDB database for storing attendance data. This architecture ensures scalability, maintainability, and efficient data management.

Teachers can monitor attendance in real-time, generate comprehensive reports, and analyze attendance statistics to identify patterns and trends. Students can check their attendance percentage, view their attendance history, and track their academic eligibility based on attendance criteria. The system also includes features for generating various reports such as daily attendance reports, monthly summaries, and student-wise attendance analysis.

The implementation of this system demonstrates how modern web technologies can significantly improve administrative processes in educational institutions. By leveraging the power of the MERN stack (MongoDB, Express.js, React, Node.js), the SPMS provides a robust, secure, and user-friendly solution for attendance management.

Keywords: Attendance Management System, Web Application, Node.js, Express.js, MongoDB, Digital Attendance, Educational Technology

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION	1
1.1 Overview of the System	1
1.2 Need for Digital Attendance Systems	2
1.3 Scope of the Project	3
1.4 Objectives of the Project	4
CHAPTER 2: SYSTEM REQUIREMENTS	5
2.1 Hardware Requirements	5
2.2 Software Requirements	6
CHAPTER 3: EXISTING SYSTEM	7
3.1 Manual Attendance Registers	7
3.2 Spreadsheet-Based Systems	8
3.3 Limitations of Existing Systems	9
CHAPTER 4: PROPOSED SYSTEM	10
4.1 System Overview	10
4.2 Advantages of Proposed System	11
CHAPTER 5: SYSTEM ARCHITECTURE	12
5.1 Three-Tier Architecture	12
5.2 System Components	13
CHAPTER 6: TECHNOLOGIES USED	15
6.1 Frontend Technologies	15
6.2 Backend Technologies	17
6.3 Database Technology	19
CHAPTER 7: DATABASE DESIGN	20
7.1 Database Collections	20
7.2 Entity-Relationship Diagram	21
7.3 Data Flow Diagram	22

CHAPTER 8: SYSTEM IMPLEMENTATION	23
8.1 Teacher Module	23
8.2 Student Module	25
8.3 Authentication System	27
CHAPTER 9: SYSTEM OUTPUT	28
9.1 User Interface Screenshots	28
CHAPTER 10: FUTURE ENHANCEMENTS	33
CHAPTER 11: CONCLUSION	35
REFERENCES	37

CHAPTER 1

INTRODUCTION

1.1 Overview of the System

The Smart Presence Monitoring System (SPMS) is a comprehensive digital attendance management system designed specifically to address the challenges faced by educational institutions in tracking and managing student attendance. In today's fast-paced educational environment, traditional methods of attendance tracking have become increasingly inadequate, consuming valuable instructional time and prone to various errors and malpractices.

The SPMS system represents a paradigm shift in how attendance is recorded, managed, and analyzed in educational settings. By leveraging modern web technologies and following industry best practices in software development, the system provides a robust, scalable, and user-friendly solution that benefits both educators and students.

At its core, the system enables teachers to create attendance sessions for their classes, generating unique session identifiers and secure PIN codes that students use to mark their attendance. This approach ensures that only students physically present in the classroom can mark their attendance, effectively eliminating the possibility of proxy attendance.

The system is built using the MERN stack architecture, which includes MongoDB as the database, Express.js as the web application framework, React for the frontend (implemented here with HTML/CSS/JS), and Node.js as the runtime environment. This technology stack was chosen for its scalability, performance, and the extensive community support available.

The SPMS is designed to be accessible from any device with a web browser and internet connection, making it convenient for users to access the system from computers, tablets, or smartphones. The responsive design ensures that the interface adapts to different screen sizes, providing an optimal user experience across all devices.

1.2 Need for Digital Attendance Systems

Traditional attendance systems used in educational institutions have several inherent limitations that affect their efficiency and reliability. Understanding these limitations is crucial to appreciating the value that digital attendance systems like SPMS bring to the educational ecosystem.

The most significant limitation of traditional attendance systems is the time consumed during the attendance marking process. In a typical classroom setting, calling out student names and waiting for responses can take anywhere from 5 to 10 minutes per session. In institutions with multiple sessions per day, this translates to a significant loss of instructional time over the course of an academic year.

Another major concern with manual attendance systems is the possibility of proxy attendance, where one student marks attendance for another who is not present. This practice undermines the integrity of attendance records and can lead to inaccurate academic assessments. Traditional systems offer no reliable mechanism to prevent such malpractices.

Human errors in attendance recording and calculation are common in manual systems. Teachers may inadvertently mark the wrong student as present or absent, or make errors when calculating attendance percentages at the end of the term. These errors can have serious consequences for students, potentially affecting their academic standing and eligibility for examinations.

Generating attendance reports from manual records is a time-consuming and error-prone process. Teachers must manually compile attendance data, calculate percentages, and prepare reports, which can take several hours for large classes. This administrative burden takes time away from more productive activities such as lesson planning and student mentoring.

Digital attendance systems address these issues by automating the entire attendance management process. The SPMS system eliminates the need for manual roll calls, prevents proxy attendance through secure PIN-based authentication, automatically calculates attendance statistics, and generates comprehensive reports with just a few clicks.

1.3 Scope of the Project

The scope of the Smart Presence Monitoring System encompasses a comprehensive set of features designed to address the attendance management needs of educational institutions. The system is designed to be flexible and adaptable to different institutional requirements while maintaining a core set of essential functionalities.

The primary scope of the system includes attendance session creation by teachers, where educators can create new attendance sessions for their classes, specify the subject, date, and other relevant details. Each session is assigned a unique identifier and a secure PIN code that students must enter to mark their attendance.

Secure attendance marking by students is another core feature within the system scope. Students can access the attendance marking interface using their credentials, enter the session ID and PIN provided by their teacher, and mark their attendance. The system records the timestamp of each attendance entry for audit purposes.

The system includes robust attendance data storage capabilities, utilizing MongoDB to store all attendance records securely. The database design ensures data integrity, supports efficient querying, and enables the generation of various reports and analytics.

Real-time attendance monitoring allows teachers to track attendance as students mark their presence. The teacher dashboard displays the current attendance status, showing which students have marked their attendance and which are yet to do so. This feature helps teachers identify and follow up with absent students promptly.

Attendance analytics and reporting capabilities provide valuable insights into attendance patterns. Teachers can generate various reports including daily attendance summaries, student-wise attendance statistics, subject-wise attendance analysis, and trend reports over specified time periods.

The system is designed to support degree programs such as B.Com, B.Sc, and B.A., with the flexibility to accommodate other programs as needed. The modular architecture allows for future expansion to include additional features and support for different educational levels.

1.4 Objectives of the Project

The Smart Presence Monitoring System was developed with the following specific objectives:

- To develop a web-based attendance management system that automates the attendance recording process, reducing the time and effort required for manual attendance tracking.
- To implement secure authentication mechanisms that prevent proxy attendance and ensure the integrity of attendance records.
- To provide teachers with real-time attendance monitoring capabilities, enabling them to track student presence during class sessions.
- To create a comprehensive reporting system that generates various attendance reports including daily summaries, student-wise statistics, and trend analysis.
- To develop a user-friendly interface that is accessible to users with varying levels of technical expertise.
- To implement a scalable database architecture that can handle large volumes of attendance data efficiently.
- To ensure data security and privacy by implementing appropriate access controls and data protection measures.
- To provide students with access to their attendance records, enabling them to monitor their attendance percentage and academic eligibility.

CHAPTER 2

SYSTEM REQUIREMENTS

2.1 Hardware Requirements

The Smart Presence Monitoring System is designed to run on standard computing hardware available in most educational institutions. The hardware requirements are minimal, making the system accessible to institutions with limited technical infrastructure.

The following hardware specifications are recommended for optimal system performance:

Component	Minimum Requirement	Recommended
Processor	Intel Core i3 or equivalent	Intel Core i5 or higher
RAM	4 GB	8 GB or higher
Storage	10 GB free space	20 GB free space
Display	1366 x 768 resolution	1920 x 1080 resolution
Network	Internet connection (2 Mbps)	Broadband (5 Mbps or higher)
Input Device	Keyboard and Mouse	Keyboard, Mouse, Touchscreen

Table 2.1: Hardware Requirements

The system can be accessed from any device with a modern web browser, including desktop computers, laptops, tablets, and smartphones. This flexibility allows users to access the system from various devices based on their convenience and availability.

2.2 Software Requirements

The SPMS system is built using modern web technologies and requires specific software components for development, deployment, and usage. The software requirements are categorized into development environment requirements and end-user requirements.

2.2.1 Development Environment

Software	Version	Purpose
Operating System	Windows 10/11, Linux, macOS	Development platform
Node.js	v18.x or higher	JavaScript runtime
MongoDB	v6.x or higher	Database server
Visual Studio Code	Latest version	Code editor
Git	Latest version	Version control
npm	v9.x or higher	Package manager

Table 2.2: Development Environment Requirements

2.2.2 End-User Requirements

Software	Version/Type	Purpose
Web Browser	Chrome v90+, Firefox v88+, Safari v14+, Edge v90+	Access web application
Operating System	Any OS with browser support	System platform
Internet Connection	Stable connection required	Server communication
PDF Reader	Adobe Acrobat or equivalent	View generated reports

Table 2.3: End-User Requirements

The system is designed to be compatible with all modern web browsers that support HTML5, CSS3, and JavaScript ES6+. Users are recommended to keep their browsers updated to the latest version for the best experience and security.

CHAPTER 3

EXISTING SYSTEM

Before the development of the Smart Presence Monitoring System, educational institutions relied on various methods for tracking student attendance. Understanding these existing systems is essential for appreciating the improvements and innovations introduced by SPMS.

3.1 Manual Attendance Registers

The most common traditional attendance system involves the use of physical attendance registers. In this system, teachers maintain paper-based registers where they manually record student attendance during each class session. The register typically contains a list of all enrolled students, and the teacher marks each student as present or absent.

The process begins with the teacher calling out student names or roll numbers, waiting for students to respond, and then marking their attendance in the register. This process is repeated for every class session, consuming significant instructional time. At the end of each month or academic term, teachers must manually calculate attendance percentages and prepare reports.

While this system is simple and requires no technical infrastructure, it suffers from numerous drawbacks. The time consumed for attendance marking reduces the effective teaching time available. Human errors in marking and calculation are common, and the system offers no protection against proxy attendance. Additionally, storing and retrieving historical attendance data from paper registers is cumbersome and inefficient.

3.2 Spreadsheet-Based Systems

Some institutions have adopted digital solutions using spreadsheet applications such as Microsoft Excel or Google Sheets for attendance management. In these systems, teachers maintain attendance records in electronic spreadsheets, which offer some advantages over paper-based systems.

Spreadsheet-based systems allow for easier data entry and automatic calculation of attendance percentages using formulas. Teachers can create templates with student information and use them to record attendance for multiple sessions. The data can be organized in various formats, and basic reports can be generated using spreadsheet features.

However, these systems still require manual data entry for each attendance session, and there is no mechanism to prevent proxy attendance. Multiple teachers cannot easily access and update the same spreadsheet simultaneously, leading to coordination challenges. Data security is also a concern, as spreadsheets can be easily copied, modified, or deleted without proper authorization controls.

3.3 Limitations of Existing Systems

The existing attendance systems suffer from several common limitations:

- **Time Consumption:** Both manual and spreadsheet-based systems require significant time for attendance marking and report generation, reducing the time available for actual teaching and learning activities.
- **Human Errors:** Manual data entry is prone to errors in marking attendance, calculating percentages, and preparing reports. These errors can affect student academic records and eligibility assessments.
- **Proxy Attendance:** Traditional systems offer no reliable mechanism to verify the identity of the person marking attendance, making proxy attendance a common problem.
- **Data Security:** Paper registers can be lost, damaged, or accessed by unauthorized persons. Spreadsheets lack robust access controls and audit trails.
- **Limited Analytics:** Existing systems provide limited capabilities for analyzing attendance patterns and trends, making it difficult to identify at-risk students or systemic issues.
- **Accessibility:** Paper registers are not easily accessible for remote monitoring or parent communication. Spreadsheets may not be accessible from all devices or locations.
- **Scalability:** As the number of students and classes increases, traditional systems become increasingly difficult to manage efficiently.
- **Integration:** Existing systems typically operate in isolation, without integration with other institutional systems such as student information systems or learning management systems.

CHAPTER 4

PROPOSED SYSTEM

The Smart Presence Monitoring System (SPMS) is proposed as a comprehensive solution to address the limitations of existing attendance management systems. The proposed system leverages modern web technologies to provide an efficient, secure, and user-friendly attendance management platform.

4.1 System Overview

The SPMS is a web-based application that enables educational institutions to manage student attendance efficiently. The system follows a three-tier architecture consisting of a presentation layer (frontend), an application layer (backend), and a data layer (database). This architecture ensures separation of concerns, making the system maintainable and scalable.

The system supports two primary user roles: Teachers and Students. Teachers have access to features for creating attendance sessions, monitoring attendance in real-time, generating reports, and managing their account settings. Students can view their attendance records, mark attendance using session credentials, and track their attendance statistics.

The core workflow of the system involves teachers creating attendance sessions for their classes. Each session is assigned a unique session ID and a secure 4-digit PIN code. Students access the attendance marking interface, enter the session ID and PIN provided by their teacher, and mark their attendance. The system records the attendance with a timestamp and updates the attendance records in real-time.

The authentication system ensures that only authorized users can access the system. Teachers and students have separate login credentials, and the system validates these credentials against the database before granting access. Password security is maintained using hashing techniques to protect user information.

4.2 Advantages of Proposed System

The Smart Presence Monitoring System offers numerous advantages over traditional attendance management methods:

- **Time Efficiency:** The digital attendance process significantly reduces the time required for attendance marking. Students can mark their attendance simultaneously, and the entire process takes only a few minutes compared to the 5-10 minutes required for manual roll calls.

Smart Presence Monitoring System (SPMS)

- **Prevention of Proxy Attendance:** The secure PIN-based authentication system ensures that only students physically present in the classroom can mark their attendance. The PIN is provided by the teacher at the beginning of the session and is valid only for that specific session.
- **Real-Time Monitoring:** Teachers can monitor attendance in real-time through their dashboard, seeing which students have marked their attendance and which are yet to do so. This enables immediate follow-up with absent students.
- **Automatic Report Generation:** The system automatically calculates attendance percentages and generates various reports including daily summaries, monthly reports, and student-wise statistics. This eliminates manual calculation errors and saves significant administrative time.
- **Data Security:** All attendance data is stored securely in a MongoDB database with appropriate access controls. The system maintains an audit trail of all attendance entries, ensuring data integrity and accountability.
- **Accessibility:** Being a web-based system, SPMS can be accessed from any device with a web browser and internet connection. This enables teachers to manage attendance from their office, classroom, or home.
- **Scalability:** The system is designed to handle large volumes of data and can scale to accommodate growing institutional needs. The MongoDB database provides horizontal scalability for handling increasing numbers of students and attendance records.
- **User-Friendly Interface:** The system features an intuitive interface that requires minimal training. Both teachers and students can quickly learn to use the system effectively.
- **Cost-Effective:** Compared to biometric or RFID-based attendance systems, SPMS requires minimal hardware investment. Institutions can deploy the system using existing computers and network infrastructure.
- **Environmental Friendly:** By eliminating paper-based attendance registers, the system contributes to environmental sustainability by reducing paper consumption.

CHAPTER 5

SYSTEM ARCHITECTURE

The Smart Presence Monitoring System follows a well-structured three-tier architecture that separates the presentation, application, and data layers. This architectural approach provides several benefits including improved maintainability, scalability, and security.

5.1 Three-Tier Architecture

The three-tier architecture is a client-server architecture in which the user interface, business logic, and data storage are developed and maintained as independent modules. This separation allows for greater flexibility and easier maintenance of the system.

The Presentation Layer (Frontend) is responsible for the user interface and user experience. It handles all user interactions, displays information to users, and collects input. The SPMS frontend is built using HTML5, CSS3, and JavaScript, providing a responsive and interactive user interface that works across different devices and screen sizes.

The Application Layer (Backend) contains the business logic of the system. It processes user requests, performs calculations, enforces business rules, and coordinates between the presentation and data layers. The SPMS backend is built using Node.js and Express.js, providing a robust and scalable server-side environment.

The Data Layer (Database) is responsible for storing and retrieving data. It manages the persistence of attendance records, user information, and session data. The SPMS uses MongoDB, a NoSQL document database that provides flexibility and scalability for handling large volumes of attendance data.

Figure 5.1: SPMS Three-Tier Architecture

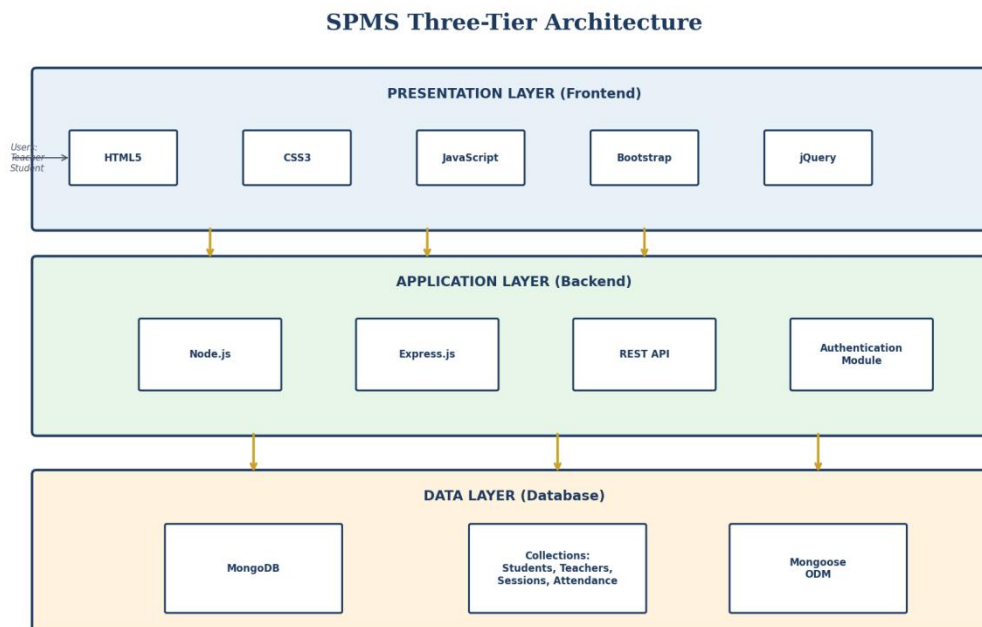


Figure 5.1: SPMS Three-Tier Architecture Diagram

5.2 System Components

5.2.1 Frontend Components

The frontend of the SPMS is built using modern web technologies that ensure a responsive and interactive user experience. The key frontend components include:

HTML5 (HyperText Markup Language): Provides the structural foundation of the web application. HTML5 semantic elements are used to create a well-structured and accessible document structure. The use of HTML5 ensures compatibility with modern browsers and provides support for advanced features.

CSS3 (Cascading Style Sheets): Handles the visual presentation and styling of the application. CSS3 features such as flexbox, grid layout, animations, and transitions are used to create an attractive and responsive user interface. Media queries enable the application to adapt to different screen sizes.

JavaScript: Provides interactivity and dynamic functionality to the application. JavaScript handles form validation, AJAX requests for server communication, DOM manipulation, and user interface enhancements. Modern ES6+ features are used for cleaner and more maintainable code.

Bootstrap Framework: A popular CSS framework that provides pre-built components and responsive grid system. Bootstrap accelerates development by providing ready-to-use UI components such as buttons, forms, navigation bars, and modals.

5.2.2 Backend Components

The backend of the SPMS is responsible for processing requests, enforcing business logic, and managing data flow. The key backend components include:

Node.js: A JavaScript runtime built on Chrome's V8 JavaScript engine. Node.js enables server-side execution of JavaScript, allowing for a unified language across the entire application stack. Its event-driven, non-blocking I/O model makes it efficient and scalable.

Express.js: A minimal and flexible Node.js web application framework. Express.js provides a robust set of features for web and mobile applications, including routing, middleware support, and template engine integration. It simplifies the creation of RESTful APIs.

REST API: The system exposes a RESTful API that allows the frontend to communicate with the backend. The API follows REST principles, using HTTP methods (GET, POST, PUT, DELETE) to perform CRUD operations on resources. JSON is used as the data interchange format.

Authentication Middleware: Implements security features including user authentication and authorization. Password hashing using bcrypt ensures secure storage of user credentials. Session management and JWT (JSON Web Tokens) can be implemented for stateless authentication.

5.2.3 Database Components

The data layer of the SPMS uses MongoDB, a document-oriented NoSQL database. MongoDB stores data in flexible, JSON-like documents, making it ideal for applications with evolving data schemas.

MongoDB provides several advantages for the SPMS including flexible schema design that accommodates changes in data structure without requiring migrations, horizontal scalability through sharding for handling large data volumes, rich query language for complex data retrieval, and built-in aggregation framework for generating reports and analytics.

Mongoose ODM (Object Document Mapper) is used to interact with MongoDB from Node.js. Mongoose provides a schema-based solution to model application data, including built-in type casting, validation, query building, and business logic hooks.

CHAPTER 6

TECHNOLOGIES USED

The development of the Smart Presence Monitoring System involves the use of various modern web technologies. This chapter provides a detailed description of the technologies used in the frontend, backend, and database components of the system.

6.1 Frontend Technologies

6.1.1 HTML5

HTML5 (HyperText Markup Language version 5) is the latest evolution of the standard markup language for creating web pages. HTML5 introduces new semantic elements such as header, nav, article, section, and footer that provide better document structure and accessibility.

Key features of HTML5 used in SPMS include semantic elements for improved document structure and SEO, form enhancements with new input types (email, date, number) and validation attributes, local storage for client-side data persistence, and canvas and SVG support for graphics and visualizations.

The SPMS uses HTML5 to create a well-structured, accessible, and standards-compliant user interface. Semantic markup ensures that the application is accessible to users with disabilities and improves search engine optimization.

6.1.2 CSS3

CSS3 (Cascading Style Sheets version 3) is the latest standard for styling web documents. CSS3 introduces numerous features that enable advanced visual effects and responsive design.

Key CSS3 features used in SPMS include Flexbox and CSS Grid for creating flexible and responsive layouts, media queries for responsive design across different screen sizes, transitions and animations for smooth user interface interactions, custom properties (CSS variables) for maintainable styling, and box shadows and border radius for modern visual design.

The SPMS uses a custom CSS framework built on CSS3 principles, ensuring a consistent and modern visual design throughout the application. The responsive design ensures that the system works well on desktop computers, tablets, and smartphones.

6.1.3 JavaScript

JavaScript is a versatile programming language that enables interactive web pages and is an essential part of web applications. Modern JavaScript (ES6+) provides powerful features for building complex applications.

Key JavaScript features used in SPMS include ES6+ syntax including arrow functions, template literals, destructuring, and modules, asynchronous programming with Promises and `async/await` for API calls, DOM manipulation for dynamic content updates, event handling for user interactions, and form validation for data integrity.

JavaScript powers the interactivity of the SPMS frontend, handling user interactions, communicating with the backend API, and dynamically updating the user interface based on data received from the server.

6.2 Backend Technologies

6.2.1 Node.js

Node.js is an open-source, cross-platform JavaScript runtime environment that executes JavaScript code outside a web browser. Node.js enables developers to use JavaScript for server-side scripting, running scripts server-side to produce dynamic web page content before the page is sent to the user's web browser.

Key features of Node.js include event-driven, non-blocking I/O model that makes it lightweight and efficient, npm (Node Package Manager) providing access to thousands of reusable packages, single-threaded event loop architecture for handling concurrent connections, and ability to handle thousands of simultaneous connections with high throughput.

Node.js is particularly well-suited for building real-time applications, APIs, and microservices. Its non-blocking I/O model makes it ideal for I/O-intensive applications like the SPMS, which involves frequent database operations and API calls.

6.2.2 Express.js

Express.js is a fast, unopinionated, minimalist web framework for Node.js. It provides a robust set of features for web and mobile applications, making it one of the most popular frameworks for building web applications and APIs.

Key features of Express.js include routing for defining application endpoints and handling HTTP requests, middleware support for adding functionality to the request-response cycle, template engine integration for server-side rendering, and error handling mechanisms for robust applications.

Express.js simplifies the development of the SPMS backend by providing a clean and intuitive API for defining routes, handling requests, and managing middleware. The framework's flexibility allows developers to structure the application according to their preferences and requirements.

6.3 Database Technology

6.3.1 MongoDB

MongoDB is a document-oriented NoSQL database used for high volume data storage. Instead of using tables and rows as in traditional relational databases, MongoDB makes use of collections and documents. Documents consist of key-value pairs which are the basic unit of data in MongoDB.

Key features of MongoDB include flexible document model that allows for varied and evolving data structures, horizontal scalability through sharding for handling large data volumes, rich query language supporting complex queries and aggregations, and high performance with indexing and in-memory operations.

MongoDB is well-suited for the SPMS because of its flexible schema that accommodates changes in data requirements, ability to handle large volumes of attendance records efficiently, and powerful aggregation framework for generating attendance reports and analytics.

6.3.2 Mongoose ODM

Mongoose is an Object Data Modeling (ODM) library for MongoDB and Node.js. It manages relationships between data, provides schema validation, and is used to translate between objects in code and the representation of those objects in MongoDB.

Key features of Mongoose include schema definition for data structure and validation, middleware (pre and post hooks) for adding logic before or after operations, query building with a fluent API, and population for referencing documents across collections.

Mongoose simplifies database operations in the SPMS by providing a structured way to define data models, validate data before saving, and perform complex queries. The middleware features enable the implementation of business logic such as password hashing and audit logging.

6.3.3 Technology Stack Summary

Layer	Technology	Version	Purpose
-------	------------	---------	---------

Smart Presence Monitoring System (SPMS)

Frontend	HTML5	Latest	Document structure
Frontend	CSS3	Latest	Styling and layout
Frontend	JavaScript	ES6+	Interactivity
Frontend	Bootstrap	v5.x	UI components
Backend	Node.js	v18.x+	Runtime environment
Backend	Express.js	v4.x	Web framework
Database	MongoDB	v6.x+	Data storage
Database	Mongoose	v7.x	ODM library

Table 6.1: Technology Stack Summary

CHAPTER 7

DATABASE DESIGN

The database design is a critical component of the Smart Presence Monitoring System. A well-designed database ensures data integrity, supports efficient queries, and enables the system to scale as the volume of data grows. This chapter describes the database structure, entity relationships, and data flow within the system.

7.1 Database Collections

The SPMS uses MongoDB as its database, which stores data in collections of documents. The following collections are defined in the system:

7.1.1 Students Collection

The Students collection stores information about registered students in the system. Each document in this collection represents a student user with the following fields:

Field	Type	Description
student_id	ObjectId	Unique identifier for the student
name	String	Full name of the student
email	String	Email address (unique)
roll_number	String	Student roll number
department	String	Department/Class of the student
password	String	Hashed password
created_at	Date	Account creation timestamp

Table 7.1: Students Collection Schema

7.1.2 Teachers Collection

The Teachers collection stores information about registered teachers. Each document represents a teacher user with the following fields:

Field	Type	Description
teacher_id	ObjectId	Unique identifier for the teacher
name	String	Full name of the teacher
email	String	Email address (unique)
department	String	Department of the teacher
password	String	Hashed password
created_at	Date	Account creation timestamp

Table 7.2: Teachers Collection Schema

7.1.3 Sessions Collection

The Sessions collection stores information about attendance sessions created by teachers. Each document represents an attendance session:

Field	Type	Description
session_id	ObjectId	Unique identifier for the session
teacher_id	ObjectId	Reference to the teacher
subject	String	Subject name
date	Date	Session date

Smart Presence Monitoring System (SPMS)

pin	String	4-digit attendance PIN
status	String	Session status (active/closed)
created_at	Date	Session creation timestamp

Table 7.3: Sessions Collection Schema

7.1.4 Attendance Records Collection

The Attendance Records collection stores the actual attendance entries marked by students:

Field	Type	Description
attendance_id	ObjectId	Unique identifier for the record
session_id	ObjectId	Reference to the session
student_id	ObjectId	Reference to the student
timestamp	Date	Attendance marking time
status	String	Attendance status (present/absent)

Table 7.4: Attendance Records Collection Schema

7.2 Entity-Relationship Diagram

The Entity-Relationship (ER) diagram illustrates the relationships between the different entities in the SPMS database. The diagram shows how students, teachers, sessions, and attendance records are related to each other.

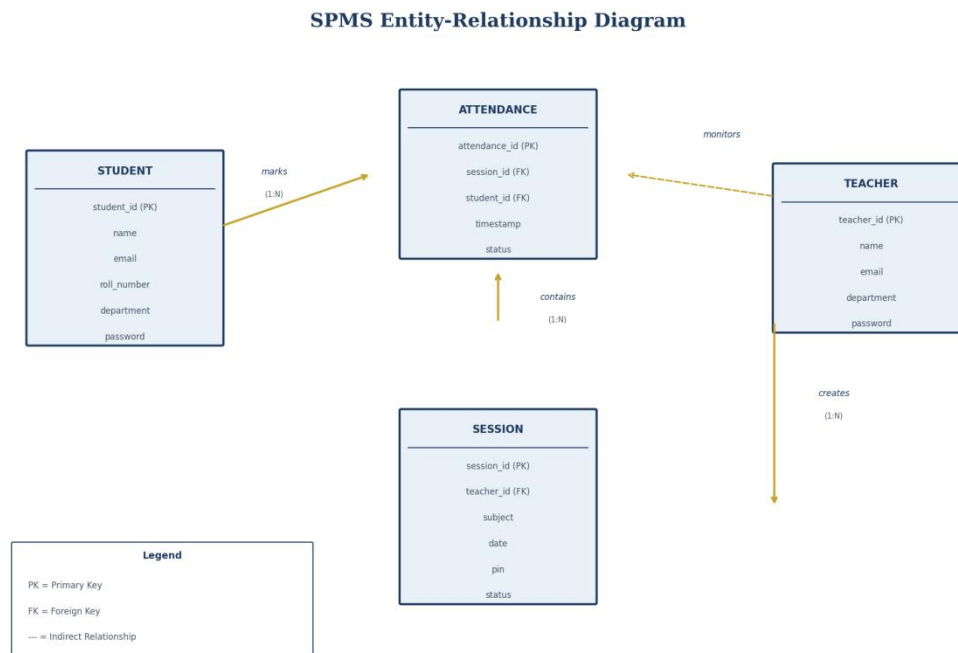


Figure 7.1: Entity-Relationship Diagram

The ER diagram shows the following relationships: A Teacher can create multiple Sessions (1:N relationship), a Session can contain multiple Attendance Records (1:N relationship), a Student can have multiple Attendance Records (1:N relationship), and a Teacher monitors Attendance Records through Sessions (indirect relationship).

7.3 Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data moves through the SPMS system. It shows the flow of information between external entities (teachers and students), processes, and data stores.

Smart Presence Monitoring System (SPMS)

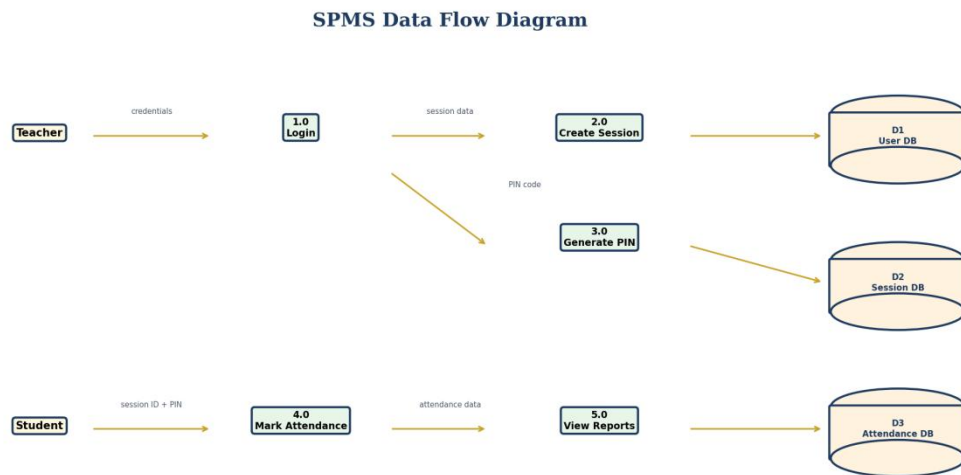


Figure 7.2: Data Flow Diagram

The DFD shows the main processes in the system: Login process handles user authentication, Create Session process allows teachers to create attendance sessions, Generate PIN process creates secure PIN codes for sessions, Mark Attendance process enables students to mark their attendance, and View Reports process generates attendance reports and statistics.

CHAPTER 8

SYSTEM IMPLEMENTATION

The implementation of the Smart Presence Monitoring System involves the development of various modules that work together to provide the complete functionality of the system. This chapter describes the implementation details of the key modules in the system.

8.1 Teacher Module

The Teacher Module provides functionality for teachers to manage attendance sessions and monitor student attendance. This module is accessible only to users with teacher credentials.

8.1.1 Teacher Login

The login functionality allows teachers to access the system using their registered email address and password. The login process involves the following steps:

The teacher enters their email address and password on the login page. The frontend validates the input format and sends the credentials to the backend API. The backend retrieves the user record from the database and verifies the password using bcrypt comparison. If the credentials are valid, a session is created and the teacher is redirected to the dashboard. If invalid, an appropriate error message is displayed.

Security measures implemented include password hashing using bcrypt with salt rounds, input validation to prevent injection attacks, rate limiting to prevent brute force attacks, and HTTPS for secure data transmission.

8.1.2 Create Attendance Session

The session creation functionality allows teachers to create new attendance sessions for their classes. The process involves:

The teacher navigates to the 'Create Session' page from the dashboard. The teacher enters the subject name and selects the date for the session. The system generates a unique session ID and a random 4-digit PIN code. The session details are saved to the database with status set to 'active'. The session ID and PIN are displayed to the teacher for sharing with students.

The PIN generation uses a cryptographically secure random number generator to ensure unpredictability. Each PIN is unique to the session and expires when the session is closed.

8.1.3 Monitor Attendance

The attendance monitoring feature provides real-time visibility into student attendance during active sessions. Teachers can view:

The total number of students enrolled in the class, the number of students who have marked their attendance, the list of present students with timestamps, the list of absent students, and attendance percentage in real-time. The monitoring dashboard auto-refreshes to show the latest attendance status without requiring manual page reloads.

8.1.4 Generate Reports

The reporting functionality allows teachers to generate various attendance reports. Available report types include:

Daily Attendance Report showing attendance for a specific date, Student-wise Attendance Report showing attendance history for individual students, Subject-wise Attendance Report aggregating attendance by subject, Monthly Summary Report providing attendance statistics for a month, and Defaulter List identifying students with low attendance percentages. Reports can be exported in PDF or Excel format for sharing and record-keeping.

8.2 Student Module

The Student Module provides functionality for students to view their attendance records and mark their attendance for active sessions.

8.2.1 Student Login

Students access the system using their registered email address and password. The login process is similar to the teacher login, with appropriate redirection to the student dashboard upon successful authentication.

8.2.2 Mark Attendance

The attendance marking process involves the following steps:

The student navigates to the 'Mark Attendance' page. The student enters the session ID provided by the teacher. The student enters the 4-digit PIN code announced in class. The system validates the session ID and PIN combination. If valid, the attendance is recorded with the current timestamp. A confirmation message is displayed to the student.

Smart Presence Monitoring System (SPMS)

Security measures include PIN validation to ensure only students in the classroom can mark attendance, timestamp recording for audit purposes, and prevention of duplicate attendance marking for the same session.

8.2.3 View Attendance Statistics

Students can view their attendance statistics through the dashboard. The statistics include:

Overall attendance percentage across all subjects, subject-wise attendance breakdown, attendance trend over time (graphical representation), number of present and absent days, and academic eligibility status based on attendance criteria. This information helps students monitor their attendance and take corrective action if needed.

8.3 Authentication System

The authentication system is a critical component that ensures only authorized users can access the system. The implementation includes:

- **Password Hashing:** User passwords are hashed using bcrypt before storage. Bcrypt automatically handles salt generation and applies multiple rounds of hashing for security.
- **Session Management:** Upon successful login, a session is created with a unique session identifier. Session data is stored server-side, and the session ID is sent to the client as a cookie.
- **Role-Based Access Control:** The system distinguishes between teacher and student roles, restricting access to features based on the user's role.
- **Input Validation:** All user inputs are validated on both client-side and server-side to prevent injection attacks and ensure data integrity.
- **HTTPS:** The system uses HTTPS for all communications, ensuring that data transmitted between the client and server is encrypted.

CHAPTER 9

SYSTEM OUTPUT

This chapter presents the user interface of the Smart Presence Monitoring System. The screenshots demonstrate the various pages and features of the system from both teacher and student perspectives.

9.1 User Interface Screenshots

9.1.1 Role Selection Page

The Role Selection page is the entry point of the system, allowing users to select their role (Teacher or Student) before proceeding to the appropriate login page.

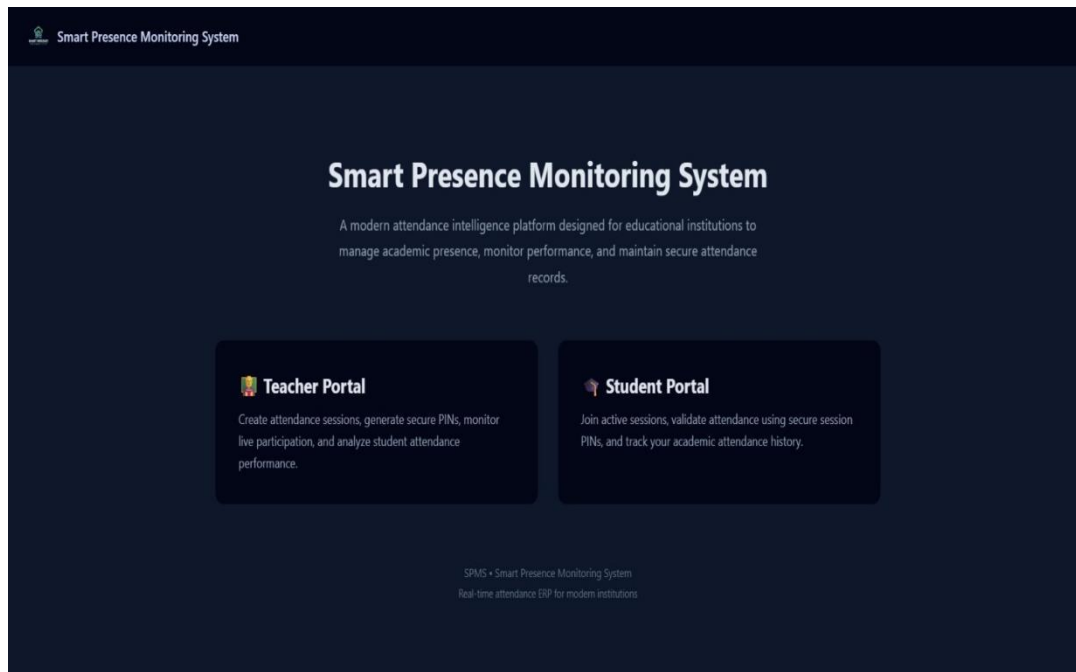


Figure 9.1: Role Selection Page

9.1.2 Teacher Login Page

The Teacher Login page allows teachers to enter their credentials to access the system. The page includes email and password fields with validation.

Smart Presence Monitoring System (SPMS)

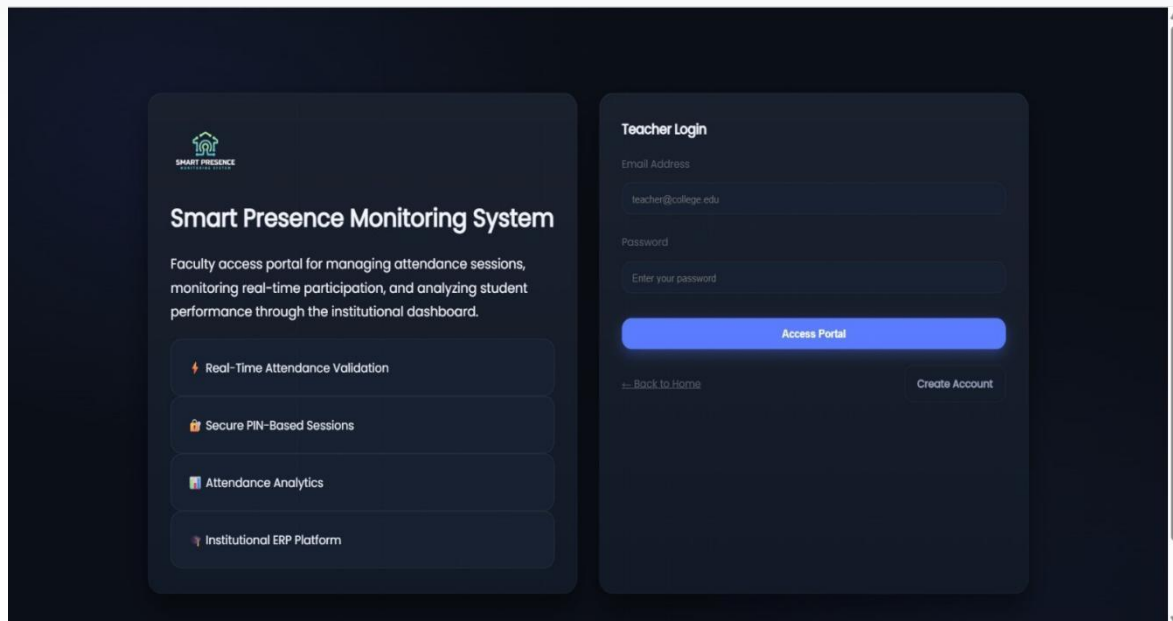


Figure 9.2: Teacher Login Page

9.1.3 Teacher Dashboard

The Teacher Dashboard provides an overview of the teacher's activities, including quick access to create sessions, view attendance, and generate reports.

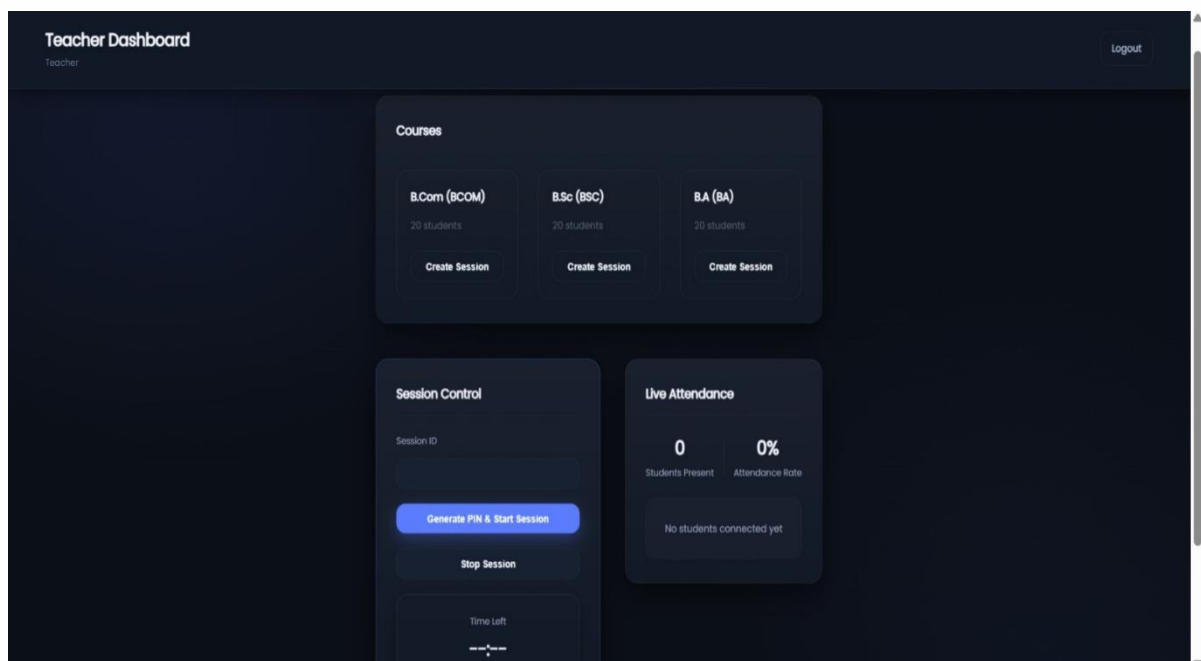


Figure 9.3: Teacher Dashboard

9.1.4 Session Creation Page

The Session Creation page allows teachers to create new attendance sessions by entering subject details and generating a secure PIN.

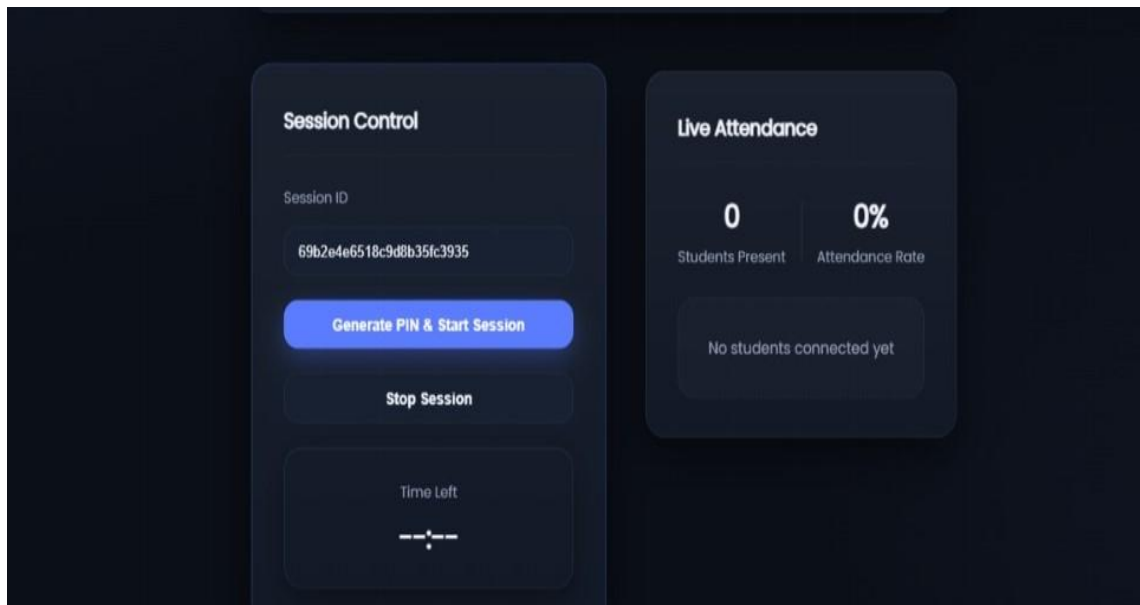


Figure 9.4: Session Creation Page

9.1.5 Student Login Page

The Student Login page allows students to enter their credentials to access their attendance records and mark attendance.

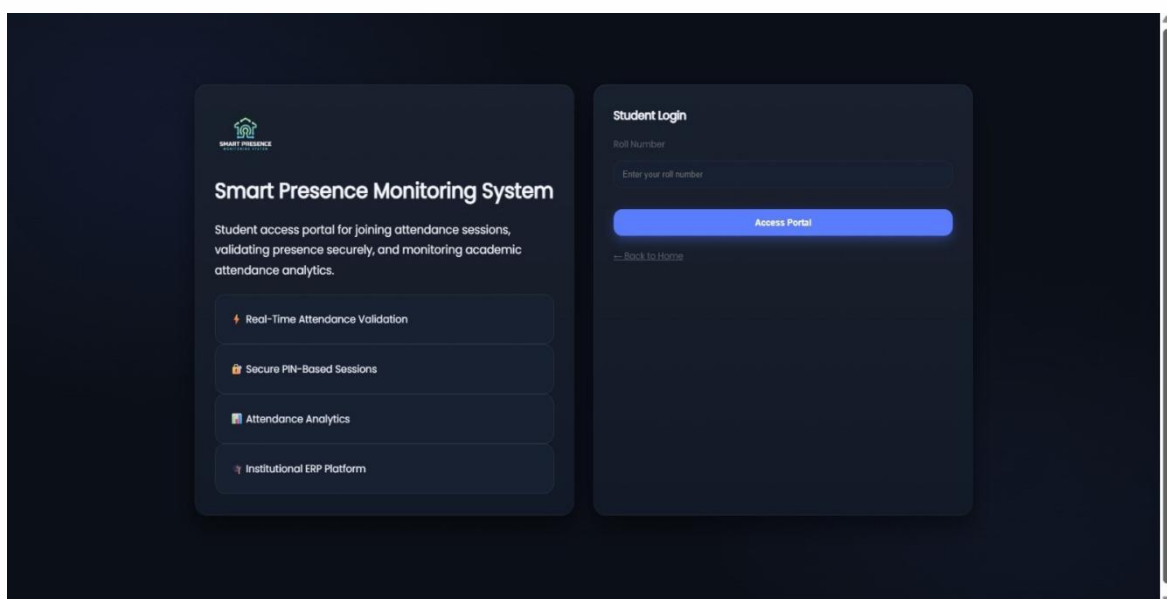


Figure 9.5: Student Login Page

9.1.6 Student Dashboard

The Student Dashboard displays the student's attendance statistics, including overall percentage, subject-wise breakdown, and recent attendance history.

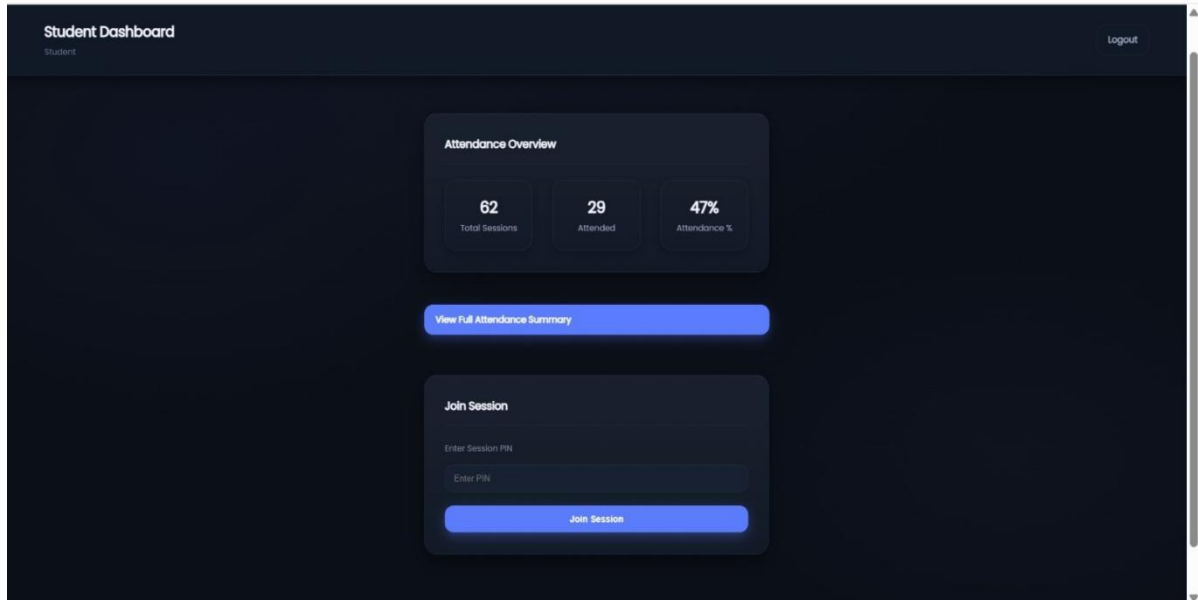


Figure 9.6: Student Dashboard

9.1.7 Attendance Marking Page

The Attendance Marking page allows students to enter the session ID and PIN to mark their attendance for an active session.

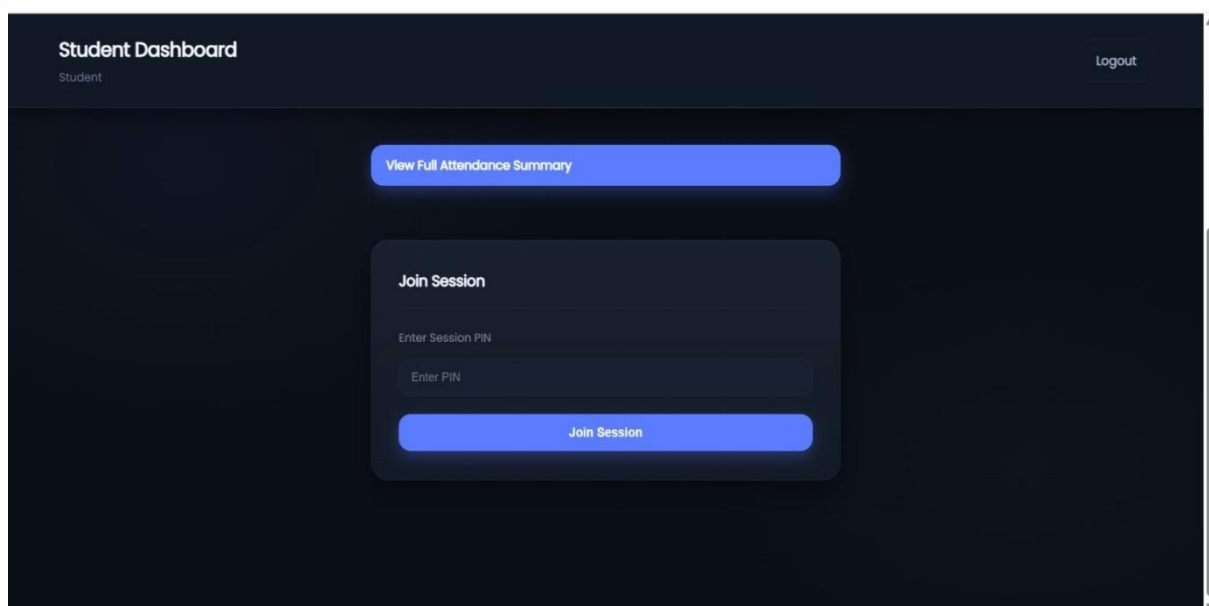


Figure 9.7: Attendance Marking Page

9.1.8 Attendance Report Page

The Attendance Report page displays detailed attendance reports with options to filter by date, subject, and export to various formats.

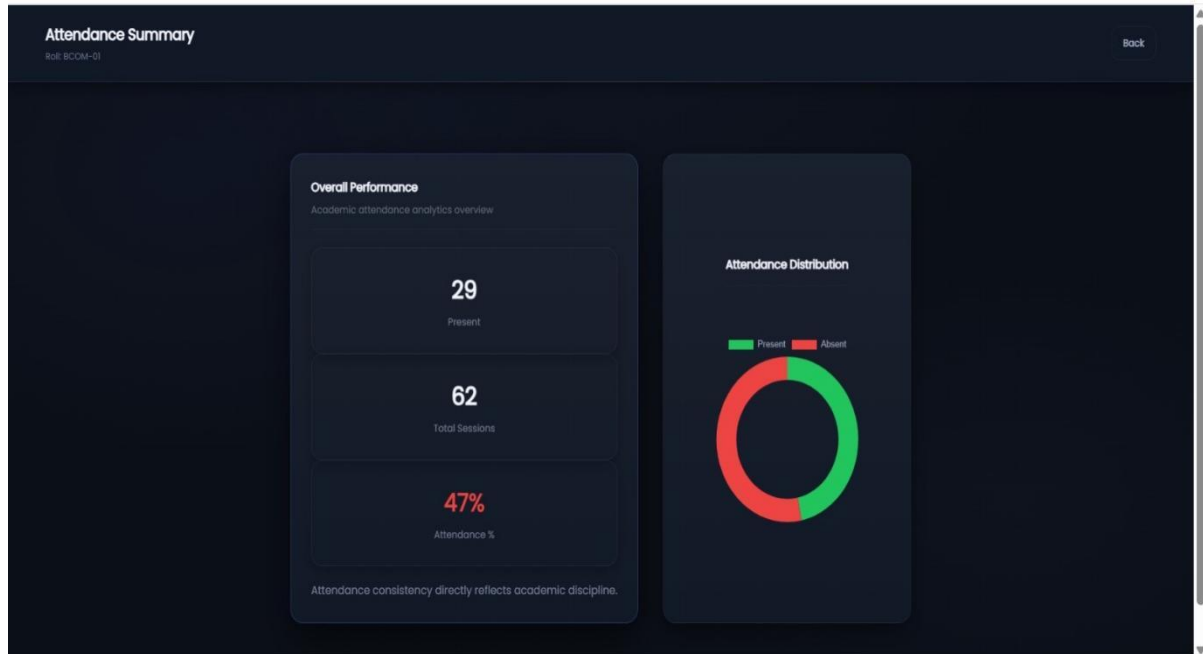


Figure 9.8: Attendance Report Page

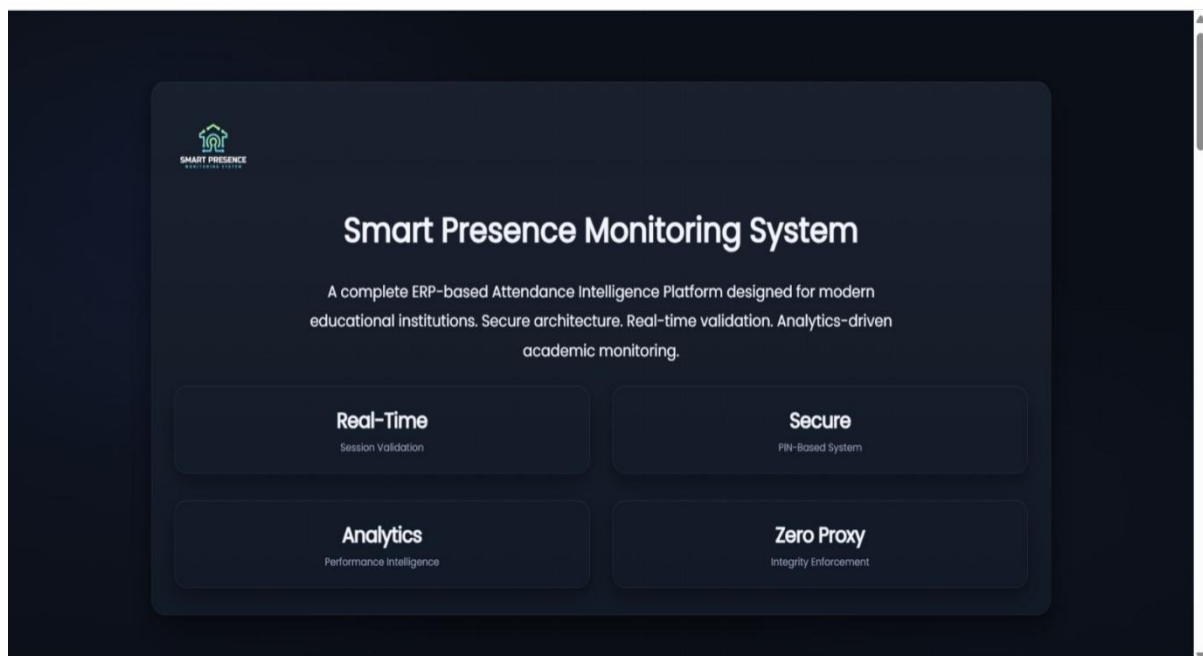
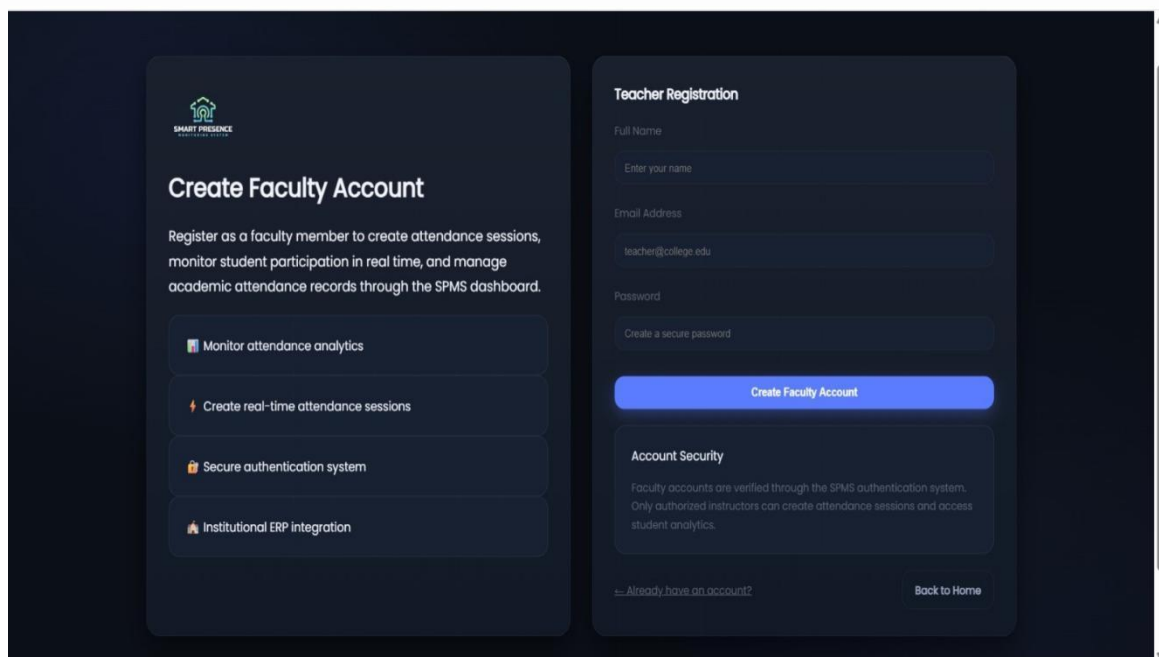


Figure 9.9: Erp Page



The image shows a 'Teacher Registration' form within the SPMS interface. On the left, a sidebar titled 'Create Faculty Account' provides instructions and lists features: 'Monitor attendance analytics', 'Create real-time attendance sessions', 'Secure authentication system', and 'Institutional ERP integration'. The main form area includes fields for 'Full Name', 'Email Address' (pre-filled with 'teacher@college.edu'), and 'Password'. A blue 'Create Faculty Account' button is prominent. Below the button, an 'Account Security' section explains the verification process. At the bottom, there are links for 'Already have an account?' and 'Back to Home'.

Teacher Registration

Full Name
Enter your name

Email Address
teacher@college.edu

Password
Create a secure password

Create Faculty Account

Account Security
Faculty accounts are verified through the SPMS authentication system. Only authorized instructors can create attendance sessions and access student analytics.

[← Already have an account?](#) [Back to Home](#)

Figure 10: Teacher Signup

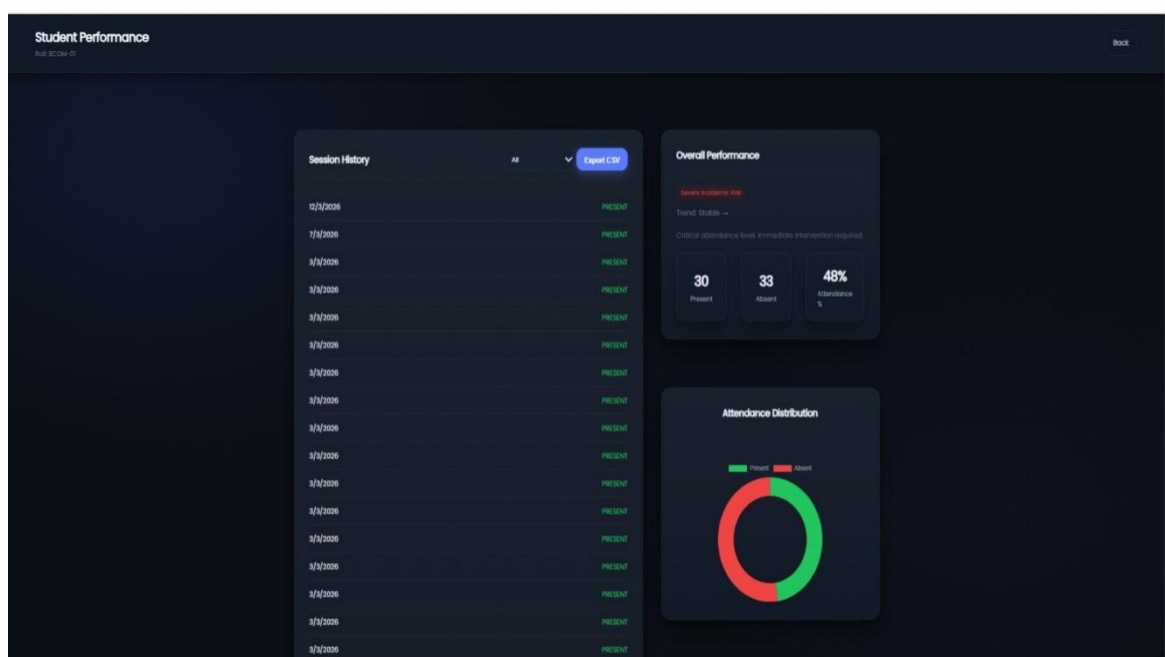


Figure 10.1: Student Performance

CHAPTER 10

FUTURE ENHANCEMENTS

While the current version of the Smart Presence Monitoring System provides comprehensive attendance management functionality, there are several enhancements that can be implemented in future versions to further improve the system.

10.1 Mobile Application Integration

A dedicated mobile application for Android and iOS platforms would provide users with a more convenient way to access the system. The mobile app could include features such as push notifications for attendance reminders, biometric authentication (fingerprint/face recognition) for secure login, offline mode for marking attendance without internet connectivity, and GPS verification to ensure students are physically present on campus.

10.2 Face Recognition Attendance

Integration of face recognition technology would provide a more secure and convenient attendance marking process. Students could mark their attendance simply by having their face scanned by a camera. This would eliminate the need for PIN codes while providing strong authentication. The system could use machine learning models to recognize students and verify their identity in real-time.

10.3 AI-Based Attendance Analysis

Artificial Intelligence and Machine Learning algorithms could be integrated to provide advanced analytics and insights. Features could include predictive analytics to identify students at risk of low attendance, pattern recognition to identify trends in absenteeism, automated alerts for unusual attendance patterns, and personalized recommendations for improving attendance.

10.4 Integration with Other Systems

The system could be integrated with other institutional systems to provide a more comprehensive solution. Possible integrations include Student Information Systems (SIS) for automatic student data synchronization, Learning Management Systems (LMS) to correlate attendance with academic performance, email and SMS gateways for automated notifications to parents and students, and calendar systems for automatic session scheduling.

10.5 Additional Features

- **Biometric Authentication:** Integration with fingerprint or iris scanners for enhanced security.
- **QR Code Attendance:** Students can scan a QR code displayed in the classroom to mark attendance.
- **Leave Management:** Students can apply for leave through the system, and teachers can approve or reject applications.
- **Parent Portal:** A dedicated portal for parents to monitor their child's attendance.
- **Multi-language Support:** Support for multiple languages to accommodate diverse student populations.
- **Dark Mode:** A dark theme option for improved user experience in low-light conditions.
- **Data Export API:** REST API for integrating attendance data with third-party applications.

CHAPTER 11

CONCLUSION

The Smart Presence Monitoring System (SPMS) represents a significant advancement in attendance management for educational institutions. By leveraging modern web technologies and following best practices in software development, the system provides a comprehensive solution that addresses the limitations of traditional attendance methods.

The system successfully achieves its primary objectives of automating the attendance recording process, preventing proxy attendance through secure PIN-based authentication, providing real-time attendance monitoring capabilities, generating comprehensive reports and analytics, and offering a user-friendly interface accessible from any device.

The three-tier architecture of the system, consisting of the presentation layer (HTML, CSS, JavaScript), application layer (Node.js, Express.js), and data layer (MongoDB), ensures scalability, maintainability, and security. The modular design allows for future enhancements and integration with other systems.

The implementation of SPMS demonstrates how technology can significantly improve administrative processes in educational institutions. By replacing manual attendance systems with a digital solution, the system saves valuable classroom time, reduces administrative burden on teachers, prevents attendance malpractices, and provides accurate attendance data for academic monitoring.

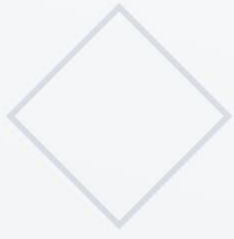
The system has been designed with flexibility in mind, allowing it to be adapted to different institutional requirements. The support for multiple degree programs and the ability to generate various types of reports make it suitable for a wide range of educational settings.

Looking ahead, the system can be further enhanced with advanced technologies such as biometric authentication, face recognition, and artificial intelligence-based analytics. These enhancements would provide even greater security, convenience, and insights for educational institutions.

In conclusion, the Smart Presence Monitoring System provides a robust, secure, and user-friendly solution for attendance management. It represents a step forward in the digital transformation of educational institutions and contributes to more efficient and effective administration of academic processes.

REFERENCES

- [1] Node.js Official Documentation. (2024). Node.js v20.x Documentation. Retrieved from <https://nodejs.org/docs/latest-v20.x/api/>
- [2] MongoDB Official Documentation. (2024). MongoDB Manual v7.0. Retrieved from <https://docs.mongodb.com/manual/>
- [3] Express.js Framework Guide. (2024). Express.js 4.x API Reference. Retrieved from <https://expressjs.com/en/4x/api.html>
- [4] Mongoose Documentation. (2024). Mongoose v8.x Documentation. Retrieved from <https://mongoosejs.com/docs/>
- [5] MDN Web Docs. (2024). HTML: HyperText Markup Language. Retrieved from <https://developer.mozilla.org/en-US/docs/Web/HTML>
- [6] MDN Web Docs. (2024). CSS: Cascading Style Sheets. Retrieved from <https://developer.mozilla.org/en-US/docs/Web/CSS>
- [7] MDN Web Docs. (2024). JavaScript Reference. Retrieved from <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
- [8] Bootstrap Documentation. (2024). Bootstrap 5 Documentation. Retrieved from <https://getbootstrap.com/docs/5.3/>
- [9] Chacon, S., & Straub, B. (2014). Pro Git (2nd ed.). Apress.
- [10] Flanagan, D. (2020). JavaScript: The Definitive Guide (7th ed.). O'Reilly Media.
- [11] Brown, E. (2019). Web Development with Node and Express: Leveraging the JavaScript Stack (2nd ed.). O'Reilly Media.
- [12] Banker, K., Bakkum, P., Verch, S., Garrett, D., & Hawkins, T. (2016). MongoDB in Action (2nd ed.). Manning Publications.
- [13] Nield, T. (2019). Getting Started with SQL: A Hands-On Approach for Beginners. O'Reilly Media.
- [14] Robbins, J. N. (2018). Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics (5th ed.). O'Reilly Media.



Smart Presence Monitoring System

(SPMS)

A Minor Project Report

Submitted by

THURPU OMKARESHWAR (111724038050)

P.SUDISH CHANDRA KUMAR (111724038045)

Academic Year 2025-2026